passes through the centre of the magnet is called magnetic neutral axis. There is no magnetic influence along this line.

Magnetic Axis (MA) is the imaginary straight line joining North to South pole. There is maximum magnetic influence along this line.

i) Magnetic Saturation

The limit beyond which the strength of magnet cannot be increased is called magnetic saturation.

ii) Residual Magnetism

It is the magnetism which remains in a material when the effective magnetizing force has been reduced to zero.

iii) Magnetic Retentivity

The property of retaining magnetism by a magnetic material is called Magnetic Retentivity.

iv) Hysteresis

The energy required to demagnetize the residual magnetism of material is known as hysteresis.

v) Leakage Flux

Leakage flux is defined as the magnetic flux which does not follow the particularly intended path in a magnetic circuit.

Taking the example, solenoid you can explain the leakage flux and the fringing both. When a current is passed through solenoid, magnetic flux is produced by it.

vi) Coercivity

Coercivity is a measure of the ability of a ferro magnetic materials to withstand an external magnetic field without becoming demagnetized.

3.4 ELECTRO MAGNETISM

When current is passed through a coil of wire, a magnetic field is set up around the coil. If soft iron bar is placed inside the coil of wire carrying current, the iron bar becomes magnetized. This process is known as electro magnetism.

The iron remains as a magnet as long as the current is flowing in the circuit. It looses its magnetism when current is switched off.

The polarity of an electromagnet depends upon the direction of the current flowing through it.

If the direction of current is altered, the polarity of the magnetic field will also be changed.

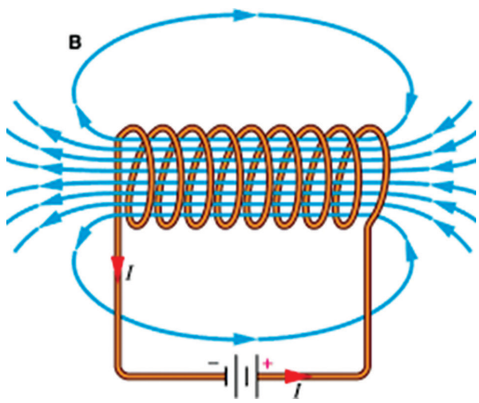


Fig 3.6 *Magnetic Field Lines*

3.4.1 Electro Magnetism in a Current Carrying Conductor

A magnetic field is formed around a conductor carrying current. The direction

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